TEXORA · LECTURE SHEET

Satin and Sateen Weaves

Satin carries the float idea to its limit: each thread interlaces only once per repeat, and the single binding points are scattered by a step (move) number chosen so that no diagonal can form. The result is the smoothest, most lustrous woven surface — and the most delicate. This sheet gives the step-number rules and works the standard 8-end construction in full.

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 4 · LEVEL 3 TERM 2

1. Definitions

A **satin** is a **warp-faced** weave in which each warp end floats over (R − 1) picks and binds under one pick per repeat of R ends × R picks, the binding point of each successive end being displaced by a constant **step (move) number S**. Because the binding points are isolated and scattered, no twill line appears and the face is an almost unbroken field of warp floats — smooth, reflective, lustrous (classically silk or filament warp). A **sateen** is the weft-faced counterpart: each **pick** floats over (R − 1) ends and binds once per repeat — classically a cotton fabric whose weft face is calendered or schreinered to imitate satin. On design paper, remember the convention: *mark = warp up* — so a satin design is mostly marked with single blanks stepping through it, while a sateen design is mostly blank with single marks stepping through it (Figures 1–3 are drawn exactly so, and confusing the two is the commonest error in this topic).

2. The step number — rules

The construction stands or falls on S. For a regular satin/sateen on repeat R:

- S must be **greater than 1** and **less than R − 1** (S = 1 or S = R − 1 steps the binding points diagonally — a twill, not a satin).
- S and R must have **no common factor** (be co-prime). If they share a factor, the binding points visit only some ends before repeating, and the remaining ends never bind in their turn — the pattern splits into sub-repeats and lines appear.

REPEAT R	VALID STEPS S	NOTE

4	none	Only 1 and 3 exist, both excluded — no regular 4-end satin (the so-called 4-end "satinette" is irregular).
5	2, 3	The smallest true satin.
6	none	2, 3, 4 all share a factor with 6; 1 and 5 excluded — no regular 6-end satin.
7	2, 3, 4, 5	7 is prime: every allowed step works.
8	3, 5	The standard worked example is S = 3 (Section 3).
9	2, 4, 5, 7	
10	3, 7	
12	5, 7	

3. Fully worked construction — 8-end satin, step 3

Rule: on pick 1, end 1 is the binding (down) end. Move S = 3 ends to the right for each successive pick; when the count passes end 8, wrap to the start (subtract 8). The binding (blank) positions are therefore:

PICK	1	2	3	4	5	6	7	8
BINDING END	1	4	7	2	5	8	3	6

Check the arithmetic line by line: 1, 1+3 = 4, 4+3 = 7, 7+3 = 10 → 10−8 = 2, 2+3 = 5, 5+3 = 8, 8+3 = 11 → 3, 3+3 = 6 — and the next would be 6+3 = 9 → 1, closing the repeat with every end visited exactly once (the co-prime rule at work). Every other square is marked: each end is up over 7 picks and down for 1, so the **warp float is 7**.

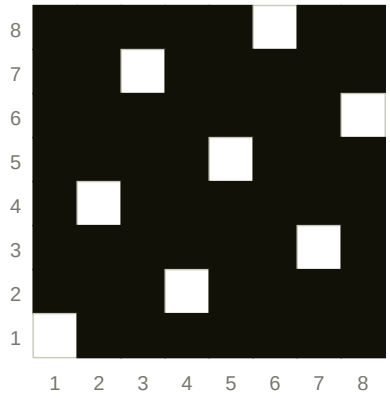


Figure 1 — 8-end satin, step 3. Marks = warp up; the single blank in each row and column is the binding point, stepping 1, 4, 7, 2, 5, 8, 3, 6. Confirm from the grid: exactly one blank per row and per column, and no two blanks adjacent diagonally in a run — hence no twill line.

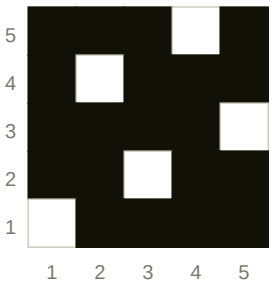


Figure 2 — 5-end satin, step 2: binding ends by pick are 1, 3, 5, 2, 4. The 5-end satin is the workhorse of cotton sateen-style warps and of satin stripes; its float is 4.

4. Sateen — the transposed construction

For a sateen, the same table is read the other way: the **marks** (warp up) are now the single binding points — one per pick and per end — because the weft must float over everything else. An R-end sateen with step S is exactly the **transpose** of the binding-point pattern of the R-end satin with step S: the binding ends of the satin table become the binding picks of the sateen.

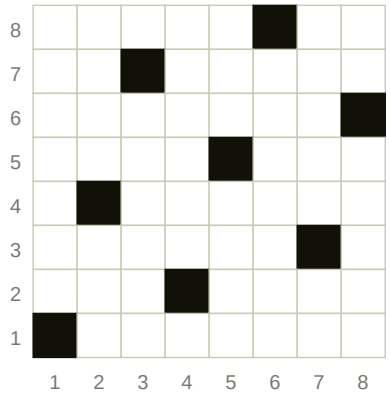


Figure 3 — 8-end sateen, step 3 (transpose of Figure 1's binding points). Each pick has just one warp end up; the weft floats over 7 ends, giving the smooth weft face. Compare cell by cell with Figure 1: the mark pattern of this grid is the blank pattern of Figure 1 turned through the leading diagonal.

5. Drafting, denting and weaving notes

- **Draft:** satin cannot use a simple straight draft economically at large repeats because each end lifts differently; ends are distributed over all R shafts in an order that evens the load — a **satin draft** (ends drawn in step order) is standard, so the peg plan becomes a simple straight sequence. For a sateen the same logic applies.
- **Denting:** commonly 2 ends per dent — pairing ends spreads the binding points evenly and hides reed marks in the float field.
- **Weaving direction:** satin is woven **face down** where possible (as a sateen arrangement on the loom) so that the long face floats do not take the friction of the reed and shuttles on the face side; the

design records must then be read inverted — a standard practical trap for the unwary.

6. Properties and uses

The float of R − 1 gives satin its virtues and defects together: **lustre and smoothness** (long parallel floats reflect light coherently), **high density potential** (few interlacings, so warp sett can be very close; satins are heavy, covered cloths), but **poor abrasion resistance and snagging** (floats are exposed and weakly anchored), **seam slippage**, and a surface that shows every weave fault. Satin: evening and bridal fabrics, linings (in sateen arrangement), satin ribbons and stripes. Sateen: cotton apparel and furnishing cloth where a smooth, lustrous weft face is wanted at cotton cost. Satin also earns its keep as a **component**: satin stripes and figures on plain grounds (Topic 5), satin-backed twills, and the rearranged orders behind crepe weaves.

> **WORKED CHECK — WHY STEP 4 FAILS ON R = 8**
>
> Binding ends with S = 4: 1, 5, 1, 5, … — only ends 1 and 5 are ever reached (4 and 8 share the factor 4). Ends 2, 3, 4, 6, 7, 8 never bind in sequence, so the "satin" collapses into separated sub-patterns with lines. This is the co-prime rule failing in public: always test S against R before drawing a single square.

Glossary

Satin — warp-faced weave, one binding per end per repeat, scattered by step S. **Sateen** — the weft-faced counterpart. **Step / move number (S)** — constant displacement of successive binding points. **Co-prime** — sharing no common factor (the condition S and R must satisfy). **Binding point** — the single interlacing of a thread per repeat.

Key points

• Satin design = full grid with single blanks stepping; sateen = its transpose with single marks stepping.
• Valid S: $1 < S < R - 1$ and $\gcd(S, R) = 1$; R = 4 and R = 6 admit no regular satin.
• 8-end satin S = 3 binds at ends 1, 4, 7, 2, 5, 8, 3, 6; float = R − 1 = 7.

Practice

1. List the binding ends, pick by pick, for a 7-end satin with step 3, and confirm the repeat closes correctly.
2. A student draws a 9-end satin with step 3. Show, in two lines, why the result is not a satin.
3. State the float length of a 10-end sateen, and which yarn system forms its face.

4. Why is a satin draft used instead of a straight draft for a 12-end satin?

Answer key

1. 1, 4, 7, 3, 6, 2, 5; next is 5+3 = 8 → 1 — every end visited once, repeat closes. (gcd(3, 7) = 1, valid.)
2. gcd(3, 9) = 3: binding ends run 1, 4, 7, 1, … — only three ends ever bind; the pattern collapses into a 3-end sub-repeat with diagonal lines.
3. Float = R − 1 = **9** ends; the face is formed by the **weft**.
4. On a straight draft each shaft would be lifted on a scattered, unequal set of picks and the lifting order is the full irregular design; the satin (step-order) draft redistributes the ends so the peg plan becomes a simple running sequence and shaft loads are balanced.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib